

SDanShare Instructions

PCB Assembly

The PCB is designed for easy assembly using cheap through hole components. The following is required.

- 1 x SDanShare PCB
- 1 x 3.3v SD Card Reader (it needs to be 3.3v connector, you cannot use the one from the Multiply which is 5v)
- 1 x 6mm Momentary Switch
- 1 x AMS1117 (these are usually SMD, but you can buy them on boards for through hole use - (which is what I have used on this to keep the soldering simple)
- 1 x SN74LVC245 (I have socketed this, but this is optional)
- 6 x Pinheaders (male)
- 6 x Cables with dupont connectors (female each end)
- 2 x LEDs (Optional)
- 2 x 330r Resistors (Only needed if using LEDs)

To connect to the Multiply to the SDanShare, remove the SD card module from the Multiply and then connect the Multiply "MicroSD" pins to the correct SD Nano pins on the SDanShare marked "Nano Connector"

Pico Firmware

The Pico needs to be flashed with the sketch is the "Pico" folder of the repository titled "SDanShare_Final_21.08.26.ino". If you are unfamiliar with flashing a Pico then there are lots of videos on Youtube that provide guidance on how to do this.

Nano Firmware

The Nano on the Multiply must be flashed using the software from Mad3003's Multiply Github, but you must make one change before doing so. The file named "Multiply_Arduino.ino" must be replaced with the modified version found in the "Nano" folder of my repository before flashing.

If you don't want to use the modified sketch and are familiar with how to amend sketches then you can just change the following in the "Multiply_Arduino.ino" file after you downloaded the code from Mad3001's github: -

Find: -

```
if (SD.begin(CS_PIN)) SdCardOk = true;
```

And change it to: -

```
if (SD.begin(CS_PIN, SD_SCK_MHZ(1))) SdCardOk = true;
```

The only functional difference between these two pieces of code is how they initialise the SD card inside the setup() function.

In order for the SDanShare to work the change forces the SD card to communicate at a much slower clock speed to maintain signal stability over certain hardware level-shifters. Cheaper or slower level-shifter chips used to translate 5V Arduino signals to 3.3V SD card signals, can distort high-frequency signals. Slowing the SPI speed to 4 MHz ensures reliable data transmission and prevents "SD Card Init Failed" errors, at the cost of slightly slower file loading times.

I have tested this with an SD card and at SPI speeds of 4 MHz file access is still quick. The only real delay is when you have folders that contain a significant number of files.

Using the SDanShare

The device is simple to use: It can be plugged in and powered as follows: -

Just plugged into PC (Pico) - PC (Pico) LED lit

Just plugged in ZX Spectrum (Nano) - Spectrum (Nano) LED lit

Plugged into both at the same time - LED lit to signify which mode you are in

When it is plugged into both, just use the button on the SDanShare to swap between modes, and the LED will tell you which mode you are in. When you switch into PC mode (Nano) the LED will pulse to show it is mounting the SD card.

If for any reason the SD card contents do not appear on the Multiply when switching, just press the reset button on the Dandanator, and the folder should appear.

The device will always default to PC mode when you plug it into a PC. So, if you are using it with your Spectrum and plug it into a PC, then it will swap to PC mode and the PC mode LED will light and pulse to show it is mounting the SD card.